

Improved Chinese few-shot relation extraction using Large Language Model for Data Augmentation and Prototypical Network

Xu Haoran

*College of Electronics
and Information Engineering
Tongji University
Shanghai, China
haoran_xu@tongji.edu.cn*

Kechen Lou

*College of Computer
Science and Technology,
Zhejiang University
of Technology
Hangzhou, China
lkc@zjut.edu.cn*

Sixian Chan*

*College of Computer
Science and Technology,
Zhejiang University
of Technology
Hangzhou, China
sxchan@zjut.edu.cn*

Abstract—Chinese few-shot relation extraction aims to effectively identify relationships between entities in text using limited data, which is crucial for information extraction and knowledge construction and assists in extracting critical information from medical cases to support personalized rehabilitation treatment plans. However, due to the limited number of samples, existing methods struggle to capture sufficient relational features from limited data, resulting in poor extraction performance. Therefore, we propose an Improved Chinese Few-Shot Relation Extraction Using Large Language Model for Data Augmentation and Prototypical Network to address this issue. Specifically, we establish the Chinese few-shot relation extraction datasets DUIE Few and SanWen Few. Notably, we introduce a framework based on large language models for dataset augmentation, which effectively alleviates the problem of feature extraction due to insufficient data and improves task performance. Finally, we present baselines for prototype networks, Siamese networks, and the CFSRE model based on relational category information. Experimental results show that the CFSRE model improves accuracy, recall, and F1 score under few-shot conditions, particularly as the sample size decreases. In summary, the method we propose demonstrates promising results in Chinese few-shot relation extraction tasks and holds the potential to advance medical rehabilitation research.

I. INTRODUCTION

Extracting entity relations from complex Chinese medical case texts has always been a research-worthy problem. Efficient extraction of entity relations can assist medical professionals in swiftly and effectively obtaining critical information from cases, enhancing their understanding of patients' conditions and treatment histories[1]. Relation extraction also facilitates establishing connections between various entities in patients' medical histories, providing support for personalized medical rehabilitation plan formulation[2, 3]. Additionally, there is a lack of relevant dataset texts for training full-shot relation extraction. Therefore, exploring how to perform Chinese relation extraction under few-shot conditions is important.

This work is partially supported by the Zhejiang Provincial Natural Science Foundation of China (Grant No. LY23F020023)

With the continuous development of natural language processing techniques, relation extraction models based on deep neural networks, such as CNN, RNN, and LSTM, have achieved excellent performance under full-shot conditions. However, these methods heavily rely on high-quality relation extraction datasets, and the cost of achieving such high-quality annotations through manual labeling is prohibitively expensive.

Therefore, there has been widespread attention to research on achieving high-quality relation extraction under few-shot conditions. The concept of few-shot learning[4] originated in computer vision, aiming to enable models to differentiate between different types by extracting features from a small number of instances, thus alleviating the pressure of high-quality annotations for unlabeled data. In recent years, especially with the rise of meta-learning, few-shot learning in the field of natural language processing has become a significant research focus. In the domain of relation extraction under few-shot conditions, models such as few-shot relation extraction based on Prototypical Networks, few-shot relation extraction based on Graph Convolutional Networks, and few-shot relation extraction models based on Siamese Networks have all demonstrated good performance.

Currently, most research models for few-shot relation extraction use the FewRel and FewRel 2.0 datasets[5, 6], both of which are English datasets. Therefore, these methods mostly focus on few-shot relation extraction in English, and there is relatively less research on few-shot relation extraction in Chinese. However, Chinese and English differ significantly in grammar, semantic expression, and writing characteristics, making it challenging to apply methods developed for one language to the other directly. Specific differences include:

- In English, the smallest unit is a word, whereas in Chinese, it is a character, which can combine to form words. This leads to more diverse ambiguity and instances of one character having multiple meanings in Chinese compared to English.
- Chinese grammar structure is more flexible, emphasizing

ing semantic expression over formal output. Sentence structures in Chinese are more loosely defined, making feature extraction for few-shot relation extraction models more challenging.

- The Chinese language evolves by combining new characters to create new words to express specific meanings based on societal needs. This phenomenon doesn't exist in English.

In November 2022, with the introduction of ChatGPT[7] by OpenAI, a revolutionary large language model, natural language processing officially entered the era of large language models. However, for few-shot relation extraction problems, the existing models, such as ChatIE[8], utilize the contextual conversational abilities of large language models but lack the utilization of their powerful information extraction and contextual understanding capabilities for information enhancement. There is also a gap in research exploring complementary approaches by combining traditional models with the strengths of large language models for few-shot relation extraction.

To address the lack of research on few-shot relation extraction in Chinese and the need to integrate traditional models with large language models, this paper fine-tunes the Baichuan-13B large language model, which performs well in the Chinese domain, to enhance its ability to output annotated data examples. The fine-tuned model is then used to augment Chinese few-shot relation extraction datasets, Few DUIE (reconstructed from the DUIE[9] full-shot relation extraction dataset proposed by Baidu) and SanWen Few (reconstructed from the SanWen[10] full-shot relation extraction dataset). A Chinese few-shot relation extraction model, CFSRE, is then trained on the augmented datasets, incorporating improvements based on the prototype network.

In summary, this paper proposes a Chinese few-shot relation extraction method that leverages the generative capabilities of large language models to mitigate data scarcity and uses label information to assist in relation extraction. The main contributions of this paper are as follows:

- We establish Chinese few-shot relation extraction datasets: DUIE Few and SanWen Few, based on the DUIE and SanWen datasets, respectively.
- We propose an improved method for Chinese few-shot relation extraction based on large language models for data augmentation. This method utilizes the powerful generative capabilities of large language models and a Rouge filtering mechanism to alleviate the difficulty of capturing relational features due to the small number of samples in few-shot conditions.
- We propose baselines for the Chinese few-shot relation extraction task, including prototype networks, Siamese networks, and a Chinese Few-Shot Relation Extraction (CFSRE) model that integrates relational category information. Experimental results indicate that under few-shot conditions, the CFSRE model shows improvements in accuracy, recall, and F1 score, with more significant

enhancements as the number of samples decreases.

II. RELATED WORK

Relation extraction, a pivotal task in natural language processing, aims to discern and classify semantic connections between entities in text, organizing them into triplets structured as <head entity, relation, tail entity>. Using the sentence “恩，中国平安，就是葛优广告那个” as an illustration, with entities <中国平安, 葛优>, a semantic analysis of the text should uncover the relationship between them as “代言人” (spokesperson), leading to the creation of a triplet instance like <中国平安, 代言人, 葛优>. Currently, research on relation extraction methods can be broadly categorized into two groups: those based on deep learning models and those based on large language models. Next, I will mainly discuss the research status of these two types of methods.

A. Relation Extraction Methods based on deep learning

In the early stages of relation extraction, most methods under full-shot conditions employed traditional machine learning approaches. For example, Che et al.[11] found that selecting the two words around entities yielded the best results in relation extraction when using Windows and SVM algorithms for feature extraction. In recent years, with the rapid development of deep neural networks such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM), they have been widely applied in the field of relation extraction. Zhou et al.[12] proposed an attention-based Bi-directional Long Short-Term Memory Network (BiLSTM), utilizing attention mechanisms to extract important information from sentences and addressing the gradient explosion problem with a bidirectional LSTM, improving the accuracy of entity relation extraction models.

B. Relation Extraction Methods based on large language model

Research on relation extraction methods based on large language models mostly focuses on designing domain-specific prompts for zero-shot and few-shot conditions to achieve domain-specific relation extraction. For example, Brinkmann et al.[14] proposed relation extraction for product text data in the product domain using ChatGPT. Experimental results indicate that ChatGPT achieves performance similar to traditional pre-trained language models under various zero-shot and few-shot conditions, with significantly less training data and fine-tuning computation required. Li et al.[15] utilized large language models like ChatGPT as zero-shot relation extractors and introduced the SUMASK prompt method combined with Conceptualization of Thought (CoT). Experimental results demonstrate that this approach enhances performance across different scales of large language models. Compared to zero-shot and fully supervised methods, it achieves competitive or superior results in relation extraction tasks.

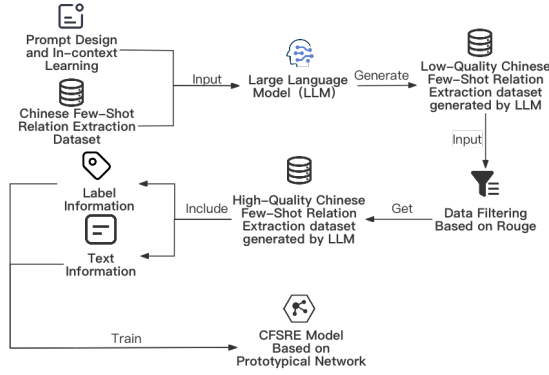


Fig. 1. Schematic diagram of the LLM-based-CFSRE model framework

In summary, currently, most relation extraction methods based on large language models convert classification problems into generation problems to achieve relation extraction, without fully utilizing the strong generative capabilities of large language models to generate training examples, thereby addressing the issue of insufficient training data for few-shot relation extraction. Therefore, we propose the ABC method to leverage the strong generative capabilities of large language models to generate additional training samples, aiming to alleviate the critical issue limiting the performance improvement of Chinese few-shot relation extraction—insufficient sample quantity.

III. METHOD

This paper proposes an improved Chinese few-shot relation extraction method based on large language models for data augmentation, implemented via the LLM-based-CFSRE model. The model comprises two modules: the first is a data augmentation module that uses a large language model, such as Baichuan-13B, fine-tuned through In-Context learning to expand the dataset and generate standardized outputs, filtered using Rouge metrics. The second part is the CFSRE module, an improvement tailored to the complexity of Chinese semantics based on the Prototypical Network model. This module uses the augmented dataset to train, aiming to optimize relation extraction performance under Chinese few-shot conditions. The overall framework of the LLM-based-CFSRE model is illustrated in Fig 1.

A. Data Augmentation Module

The data augmentation module consists of two main stages. The large language model is fine-tuned in the first stage through In-Context learning. Utilizing powerful generative capabilities of open-source large language models like Baichuan-13B, this stage generates additional formatted textual data for the few-shot dataset. The generated dataset is filtered in the second stage using text similarity metrics such as Rouge-L. This process ensures that the generated training examples not only encompass the targeted relations but also contain crucial deep features that are challenging to extract from the existing few-shot dataset.

1) *LLM-based Data Generating*: First, the prompt will be constructed to transform the task of expanding the Chinese few-shot relation extraction dataset into a language generation task. Let D represent the Chinese few-shot relation extraction dataset to be developed, where each data sample, denoted as d_i , includes entity pairs $entities = (e_{head}, e_{tail})$, Chinese sentence token, and the corresponding relation Rel . Entities contain essential entity information such as character details and position information for both the head and tail entities. The primary task in this step is to generate an enriched dataset D_{gen} based on the existing Chinese few-shot relation extraction dataset D using a language model, such as the Baichuan-13B model.

A prompt is constructed for each data sample in the Chinese few-shot relation extraction dataset. This prompt is input into the Baichuan-13B model and consists of the following components:

- *Instruction Ins* : Design a template for task generation that uses the sentence information from existing data samples to prompt the language model. The model is required to generate new sentences that include the specified entities and the relationship between them. The prompt also specifies the desired quantity of generated data samples.
- *In-Context Learning Demonstration ICL_{Demo}* : Manually construct several high-quality QA examples based on existing data samples. Save these QA examples to a QA repository $QA Rep$. For each prompt, randomly select n QA examples from $QA Rep$ as In-Context learning demonstrations, i.e., $ICL Demo = random(QA Rep, n)$.
- *Input $Input$* : The Input component includes the character information $(e_{head}_{char}, e_{tail}_{char})$ for the current data sample, the sentence information Token of the current data sample, and the specified relationship type Rel for the text to be generated.

In summary, the process of data generation based on the large language model can be represented as:

$$D_{gen} = Baichuan(I, ICL Demo, Input) \cap D \quad (1)$$

$$ICL Demo = random(QA Rep, n) \quad (2)$$

$$\{d_{iEntities}, d_{iRel}, d_{iToken}\} \in Input \quad (3)$$

2) *Rouge-based Data Filtering*: Even though large language models possess generative solid capabilities, the generated data examples may still exhibit irregular formatting and content that is too divergent, leading to instances where the required entity relationships do not exist. Therefore, performing a secondary filtering of the data text generated by large language models is necessary. In the filtering process, several factors are primarily considered:

- *Presence of required named entities*: If the required named entities are absent, it indicates that the generated sentence does not meet the requirements and that there

are no corresponding entity relationships. Texts of this nature will be directly discarded.

- Encouraging diversity in the expression of relationships in the generated text: The semantic similarity metric Rouge-L is computed between the sentences of the provided examples and the generated sentences. When the Rouge-L value exceeds 0.8, the two sentences are considered overly similar, providing little assistance to the CFSRE model. Such cases will also be discarded.
- Ensuring the generated sentences still have the required entity relationships: The semantic similarity metric Rouge-L is again utilized to maintain the entity relation in the generated sentences. For generated texts with Rouge-L values below 0.2, the ChatGPT model determines whether the entity relationships between the two sentences still exist. If the relationships exist, the sentence will be retained; otherwise, it will be discarded.

ships

Recall-Oriented Understudy for Gisting Evaluation (Rouge)[18] is a set of metrics used to automatically evaluate text generation systems. ROUGE-L utilizes the length of the longest common subsequence (LCS) as a criterion for assessing similarity. The specific calculation formula is as follows:

$$R_{lcs} = \frac{LCS(x, y)}{len(x)} \quad (4)$$

$$P_{lcs} = \frac{LCS(x, y)}{len(y)} \quad (5)$$

$$Rouge - L = \frac{2 * R_{lcs} * P_{lcs}}{R_{lcs} + P_{lcs}} \quad (6)$$

Where, $LCS(x, y)$ represents the length of the longest common subsequence between the sentences x and y , and $len(x)$ indicates the length of the sentence x being evaluated.

B. CFSRE Module

In this section, we will primarily introduce the detailed implementation process of the CFSRE module (As depicted in Fig 2). The module takes as input the N-Way-K-Shot generated dataset D_{gen} , which has been enhanced through the data augmentation module. The dataset for each task is randomly divided into two sets: a support set ($Support = S_n^k, n = 1, 2, \dots, N, k = 1, 2, \dots, K$) and a query set ($Query = Q_n^q, n = 1, 2, \dots, N, q = 1, 2, \dots, Q$). Here, K represents the number of categories to be classified, N denotes the number of support samples provided for each category, and Q is the number of query samples provided for each category. Drawing inspiration from the Prototypical Network model, Chinese-Bert-wwm is employed as the encoder to encode the input text. The module then combines the positional information of entity mentions with the semantic information of labels for few-shot Chinese relation extraction. This process determines the prototype center of each category in a high-dimensional vector space. Finally, the model computes the distance from the sample to the prototype centers of different categories to determine its relationship category.

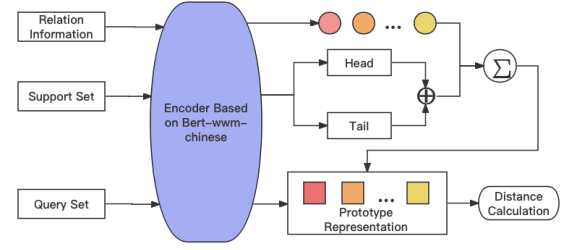


Fig. 2. Schematic diagram of the CFSRE module framework

1) *Encoding of Sentences and Relations*: In this paper, the sentence and relation encoding section utilizes the Chinese-Bert-wwm model[19] proposed by Harbin Institute of Technology and iFLYTEK. Baldini Soares et al.[20] pointed out in their 2019 work that linking the hidden states of the start tokens mentioned by two entities can better capture the status of instances in the support and query sets. Therefore, it is necessary to input texts annotated with the head $entityText_{head}$ and texts annotated with the tail $entityText_{tail}$ separately into the Chinese-Bert-wwm model for encoding in this paper. The vector representations obtained are Emb_{head} and Emb_{tail} , and these two vectors are then concatenated to obtain the vector representation of the textual sentence, denoted as Emb_{Text} . Additionally, the labels and semantic information of each type of relation are also input into the encoder to obtain the vector representation of the relation, denoted as Emb_{Rel} . This process simply utilizes the Chinese-Bert-wwm encoder to encode both the textual sentence and the relation type separately. The encoding of the textual sentence includes encoding for texts annotated with the head and tail entities.

2) *Relation Representation*: The prototype network achieves relation representation by computing the average of all samples belonging to a category, which is considered the prototype center position of that category in a high-dimensional space. Assuming the prototype center positions of the relations to be classified are denoted as $P = \{P_i \in R^{2 * h_{size}}, i = 1, 2, \dots, N\}$, where h_{size} represents the length of the vector output by the encoder, which is generally considered to be equal to the hidden layer size of the model, and $R^{2 * h_{size}}$ denotes vectors of length $2 * h_{size}$. Because of the limited number of samples, each sample is treated equally in the classification process. Therefore, the specific calculation formula for P_i is as follows:

$$P_i = \frac{1}{K} \sum_{j=1}^K Emb_n^k \quad i = 1, 2, \dots, N \quad (7)$$

$$Emb_n^k = Emb_{Text_n}^k + Emb_{Rel_n}^k \quad (8)$$

Where $+$ denotes vector addition, Emb_n^k represents the encoded result of the text data sample to be processed, which belongs to the $n - th$ category and is the $k - th$ support set example. $Emb_{Rel_n}^k \in R^{2 * h_{size}}$. Emb_n^k is formed by concatenating $Emb_{head_n}^k$ and $Emb_{tail_n}^k$ from Section

III-B1 Encoding of Sentences and Relations, and the specific calculation process is shown in the following formula:

$$Emb_{Text_n}^k = Emb_{head_n}^k \oplus Emb_{tail_n}^k \quad (9)$$

Where $\oplus$ denotes the vector concatenation operation, and $Emb_{head_n}^k, Emb_{tail_n}^k \in R^{h_{size}}$.

The relationship representation section aims to integrate the vector representations obtained by embedding text and relationships using the Chinese-Bert-wwm encoder. This integration seeks to determine the prototype center positions more accurately for the classified relationship types. During the fusion process, label features are combined with semantic features to enhance the model's generalization ability to unknown categories, thereby improving the model's accuracy in the few-shot relation extraction task. In this context, label features are introduced, serving as a guide for the model better to cluster different relationship types in the prototype space. This combination during the fusion process allows the CFSRE module to learn the commonalities of relationship types from limited samples, contributing to improving the model's performance in extracting Chinese relations under few-shot conditions.

3) *Relation classification*: In the relationship classification process, the CFSRE model uses the vector dot product to calculate the distance from the query instance Q to each prototype's center position and utilizes $max()$ to select the relationship class with the shortest distance as the classification result. The specific calculation process is shown in the following formula:

$$\begin{aligned} P_{Q_i} &= \max(P_\lambda(y = r_i | Emb_{head_n}^q)) \\ &= \max(-\exp(d(Emb_{head_n}^q, P_i))) \end{aligned} \quad (10)$$

Where $Emb_{head_n}^q$ represents the encoding result of the q -th query set instance belonging to the n -th class, $n \in 1, 2, \dots, N, q \in 1, 2, \dots, Q$. The function $d(\cdot, \cdot)$ calculates the distance between two vectors. $P_\lambda(y = r_i | Emb_{head_n}^q)$ represents the probability that the query instance $Emb_{head_n}^q$ is of relationship type r_i . Here, we choose the Euclidean distance as the distance function. Finally, we adopt the cross-entropy as the loss function, and the specific calculation process is shown in the following formula:

$$\mathcal{L}_{FSRE} = -\log Z_y \quad (11)$$

Where y is the category number, and Z_y is the estimated probability of category y .

IV. EXPERIMENTAL SETUP

A. Experimental Setup

1) *Dataset*: This paper utilized two commonly used Chinese full-shot relation extraction datasets, the DUIE Chinese relation extraction dataset proposed by Baidu and the SanWen Chinese relation extraction dataset proposed by Xu et al. The former contains 50,000 instances with 49 relation types, while the latter consists of 17,000 instances with 9 relation types. For the few-shot condition, these two datasets were

restructured to construct the N-Way-K-Shot few-shot Chinese relation extraction datasets, namely, DUIE Few (where $N = [1, 5, 10]$, $K = [1, 5, 10]$) based on DUIE, and SanWen Few (where $N = [1, 5, 9]$, $K = [1, 5, 10]$) based on SanWen.

2) *Evaluation Indicators*: This study employed commonly used metrics in relation to extraction model evaluation, namely, Precision (P), Recall (R), and F1 score (F1), to assess the model's classification results for each individual category. The specific calculation formula is as follows:

$$Pre_{avg} = \sum_{i=0}^N \frac{S_i}{SumN} Pre_i \quad (12)$$

$$Recall_{avg} = \sum_{i=0}^N \frac{S_i}{SumN} Recall_i \quad (13)$$

$$F1_{avg} = \frac{2 * Pre_{avg} * Recall_{avg}}{Pre_{avg} + Recall_{avg}} \quad (14)$$

Where, Pre_i is the precision of the i -th relation, $Recall_i$ is the recall of the i -th relation, N represents the total number of classes to be classified, S_i represents the number of samples for the i -th relation, $SumN$ represents the total number of samples for the i -th relation, Pre_{avg} , $Recall_{avg}$, and $F1_{avg}$ are used as the model's precision, recall, and F1 score, respectively.

3) *Parameter Settings*: In the experiments, the $maxlength$ is set to 50, and if the character count of a sentence exceeds 50, the first 50 characters are selected; if the character count is less than 50, it is padded with the character "[PAD]". Regarding word embeddings, this study utilizes the pre-trained Chinese-Bert-wwm model to encode input text sequences. Since this model encodes word vectors with a dimensionality of 768, the hidden size is set to 768. All models are trained on the training set, and then the model performing best on the validation set is selected for testing.

B. Experimental Results

To demonstrate the effectiveness of the proposed improved Chinese few-shot relation extraction approach based on large language models for data augmentation, comparisons were conducted with several commonly used few-shot relation extraction models on the DUIE Few and SanWen Few datasets. The achieved results are presented in Tables I and II. The baseline models used include:

- **BiLSTM**: Utilizes the pre-trained Chinese-Bert-wwm model to encode text. It then employs a BiLSTM layer to capture contextual information and sequential patterns within the sequences, understanding dependency relationships in the text. Finally, a fully connected layer is used for relation extraction classification.
- **Proto**: For each class to be classified, it calculates the average representation of the samples in its training set, obtaining a prototype in the feature space and representing the center of the samples for that class. For

TABLE I
EXPERIMENTAL RESULTS TABLE ON THE DUIE FEW DATASET FOR EACH MODEL

		BiLSTM	Proto	Siamese	HCRP	LLM-based- FSRE (Our model)
5W 1S	P	70.66	82.95	81.12	82.92	90.50
	R	64.94	71.36	69.58	70.33	83.10
	F1	67.68	76.72	74.91	76.11	86.64
5W 5S	P	87.99	91.56	89.17	91.27	92.60
	R	83.55	84.94	82.06	84.43	86.30
	F1	85.71	88.12	85.47	87.71	89.34
5W 10S	P	94.27	93.75	93.61	92.97	98.87
	R	95.93	88.88	88.83	87.50	85.46
	F1	95.09	91.25	91.16	90.15	91.68
10W 1S	P	57.93	80.29	70.35	74.59	84.69
	R	35.27	67.97	56.97	61.37	74.18
	F1	43.85	73.62	62.95	67.34	79.09
10W 5S	P	71.31	85.97	80.33	86.23	96.83
	R	78.59	76.05	70.77	76.07	77.87
	F1	74.77	80.71	75.25	80.83	86.32
10W 10S	P	85.27	87.56	86.50	91.09	97.29
	R	89.86	79.43	78.67	84.93	81.88
	F1	87.50	83.30	82.40	87.90	88.92

TABLE II
EXPERIMENTAL RESULTS TABLE ON THE SANWEN FEW DATASET FOR EACH MODEL

		BiLSTM	Proto	Siamese	HCRP	LLM-based- FSRE (Our model)
5W 1S	P	76.43	59.05	58.41	64.65	63.14
	R	17.94	43.52	43.52	47.09	49.51
	F1	29.06	50.11	49.88	54.49	55.50
5W 5S	P	76.66	72.79	72.15	77.02	78.13
	R	59.83	60.33	60.19	65.13	65.56
	F1	67.21	65.98	65.63	70.58	71.30
5W 10S	P	77.92	92.06	91.68	80.86	95.04
	R	70.02	71.60	71.24	69.60	72.86
	F1	73.76	80.55	80.18	74.81	82.48
10W 1S	P	45.07	55.11	54.60	42.62	63.50
	R	17.74	27.91	26.93	26.30	34.63
	F1	25.45	37.06	36.07	32.53	44.82
10W 5S	P	47.81	67.81	62.66	59.47	78.67
	R	38.41	43.18	36.66	41.50	47.39
	F1	42.60	52.76	46.25	48.89	59.15
10W 10S	P	50.37	78.08	73.80	64.57	86.61
	R	43.72	52.98	46.43	49.26	53.24
	F1	46.81	63.12	57.00	55.88	65.95

given test samples, it calculates the distance to each class prototype. By comparing these distances, the sample is classified into the class represented by the nearest prototype.

- **Siamese:** Comprising two identical subnetworks with shared weights, Siamese extracts features for sample pairs through shared weights. The extracted features undergo a dissimilarity measurement to obtain the similarity between sample pairs. The learning objective involves minimizing the difference between pairs of the same class and maximizing the difference between pairs of different classes, aiming to learn a measure of similarity between sample pairs.
- **HCRP:** A few-shot relation extraction method based on contrastive learning. This method utilizes relation-label information to learn better representations, enabling the model to perform better in challenging tasks.

Based on the results recorded in the experimental results table above, the following conclusions can be drawn:

- The proposed improved Chinese few-shot relation extraction approach based on large language models for data augmentation, shows improvement in accuracy, recall, and F1 score values compared to traditional few-shot methods. This improvement is particularly evident in the DUIE Few dataset, with the most significant enhancements observed in the 5Way-1Shot and 10Way-1Shot experiments, where F1 values increased by 9.92 and 5.47, respectively. This indicates that the proposed method can enhance the diversity of data samples under few-shot conditions, alleviating the issue of insufficient coverage of data samples in such conditions, leading to improved model performance.
- In Chinese relation extraction tasks with a small number of categories (e.g., 5Way tasks), the performance of pro-

totype networks and siamese networks is comparable, each with its own advantages and disadvantages. However, when the number of categories in the classification task is larger (e.g., 10Way tasks), the classification performance of prototype networks is significantly better than that of siamese networks. As the number of samples increases, the performance gap between the two methods becomes more apparent.

- The improvement method proposed in this study, based on data augmentation using large language models, achieves significant improvements in accuracy, recall, and F1 values on the simple task dataset (DUIE Few) compared to the complex task dataset (SanWen Few). This phenomenon suggests that large language models have a more in-depth understanding of simple tasks, providing more complete coverage of unforeseen situations in the output data samples. However, for diverse and content-rich texts like prose data, the coverage is not complete enough and may not encompass all possible scenarios for this category.
- In the 5Way10Shot scenario of the DUIE Few dataset, traditional full-shot methods outperform the proposed model, aligning with the findings of Ma et al.[21]. For simple tasks, such as a 5-classification problem, having 10 samples for each category is already considered information-rich. Therefore, under these conditions, few-shot relation extraction methods may not perform as well as full-shot methods. Further experiments reveal that the dataset enhanced through the data augmentation module can still improve the performance of traditional full-shot methods (see Section IV-D Data Augmentation Experiments for details).

C. Ablation Study

This study conducted the following ablation experiments to assess the impact of not using the data augmentation module based on large language models on the CFSRE model and not using the CFSRE module on the LLM-based-proto model. The experiments were performed on the DUIE Few dataset. A comparison was made with the proposed Chinese few-shot relation extraction improvement method based on large language models and traditional prototype network methods. This aimed to validate the individual and combined effects of these two modules on model performance and confirm the effectiveness of the designed modules. The specific experimental results are presented in Table III.

TABLE III
TABLE OF ABLATION EXPERIMENT RESULTS

		Proto	CFSRE	LLM-based proto	LLM-based -CFSRE
5W 1S	P	71.36	86.87	89.27	90.50
	R	82.95	77.88	82.56	83.10
	F1	76.72	82.13	85.78	86.64
5W 5S	P	91.56	91.07	91.60	92.60
	R	84.94	84.50	85.95	86.30
	F1	88.12	87.66	88.69	89.34
5W 10S	P	93.75	93.33	93.66	98.87
	R	88.88	88.21	88.86	85.46
	F1	91.25	90.70	91.19	91.68
10W 1S	P	80.29	91.08	83.06	84.69
	R	67.97	62.58	73.39	74.18
	F1	73.62	74.19	77.93	79.09
10W 5S	P	85.97	86.38	88.12	96.83
	R	76.05	77.33	80.72	77.87
	F1	80.71	81.61	84.25	86.32
10W 10S	P	87.56	90.38	92.15	97.29
	R	79.43	83.38	86.47	81.88
	F1	83.30	86.74	89.22	88.92

Through the ablation experiments, we found that both the data augmentation module based on a large language model and the CFSRE module proposed in this paper can effectively improve the performance of the model on Chinese few-shot relation extraction tasks. The experimental results of the LLM-based-proto model are significantly better than those of the CFSRE model, indicating that the data augmentation module based on a large language model has a greater impact on improving model performance compared to the CFSRE module. This also demonstrates that the main factor limiting the achievement of better performance in relation extraction tasks under few-shot conditions is the quantity of data examples. Additionally, the experimental results show that when the sample size is small (under 1Shot and 5Shot conditions), our CFSRE model performs better on the Chinese few-shot relation extraction task dataset DUIE Few than the traditional Proto model. However, when the sample size is relatively large (under 10Shot conditions), our CFSRE model does not perform as well as the Proto model, indicating that our model is more suitable for few-shot Chinese relation extraction tasks

with fewer samples. Of course, only in the LLM-based-CFSRE model, where both modules are effective, can our method achieve the best results.

D. Data Augmentation Experiments

To demonstrate that the dataset generated by the data augmentation module designed in this paper is not only effective for our proposed CFSRE model but can also improve the performance of other full-shot methods (BiLSTM) and few-shot methods (Proto and Siamese), we conducted experiments on the enhanced DUIE Few Chinese few-shot relation extraction dataset using traditional BiLSTM, prototype network, and siamese network. The specific experimental results are shown in Table IV.

TABLE IV
EXPERIMENTAL RESULTS TABLE OF ENHANCED DATASET DUIE
FEW ON VARIOUS MODELS

		BiLSTM	LLM -based BiLSTM	Proto	LLM -based Proto	Siamese	LLM -based Siamese
5W 1S	P	70.66	78.17	71.36	89.27	81.12	89.94
	R	64.94	73.26	82.95	82.56	69.58	82.88
	F1	67.68	75.64	76.72	85.78	74.91	86.26
5W 5S	P	87.99	97.81	91.56	91.60	89.17	92.36
	R	83.55	97.80	84.94	85.95	82.06	86.88
	F1	85.71	97.80	88.12	88.69	85.47	89.54
5W 10S	P	94.27	97.50	93.75	93.66	93.61	94.33
	R	95.93	97.51	88.88	88.86	88.83	90.06
	F1	95.09	97.50	91.25	91.19	91.16	92.14
10W 1S	P	57.93	68.02	80.29	83.06	70.35	77.06
	R	35.27	58.59	67.97	73.39	56.97	63.72
	F1	43.85	68.13	73.62	77.93	62.95	69.76
10W 5S	P	71.31	88.19	85.97	88.12	80.33	85.12
	R	78.59	86.84	76.05	80.72	70.77	77.04
	F1	74.77	87.51	80.71	84.25	75.25	80.88
10W 10S	P	85.27	88.94	87.56	92.15	86.5	90.02
	R	89.86	88.31	79.43	96.47	78.67	83.43
	F1	87.50	88.62	83.30	89.22	82.40	86.60

According to the experimental results in Table 5, it is evident that the dataset enriched by the Baichuan-13B large language model not only benefits the proposed CFSRE method in various sample training conditions but also demonstrates positive gains for traditional full-shot learning methods (BiLSTM) and few-shot learning methods (Proto and Siamese). Specifically, the most significant improvement is observed in the 5-Way1-Shot, where the enhanced dataset achieves respective F1 value improvements of 7.96, 9.06, and 11.35 for different models. For traditional full-shot learning methods, the performance of our method on the enhanced dataset in the 5-Shot can match or even exceed the performance of the non-enhanced dataset in the 10-Shot. Additionally, in the 5-Way10-Shot, the enhanced dataset has a negative impact on the Proto model. We speculate that this might be due to limitations inherent in the Proto model itself. For the 5-Way10-Shot task in the DUIE Few dataset of Chinese relation extraction, the F1 value performance of 91.25 may have already reached the bottleneck of the model, leading to a negative gain situation.

V. CONCLUSIONS

In this work, we restructured the Chinese few-shot relation extraction datasets DUIE Few and SanWen Few, proposing a framework that enhances these datasets through contextually fine-tuned large language models and a Rouge-based filtering mechanism to address challenges in feature extraction due to dataset limitations. Additionally, we improved the prototype network model by integrating relational category information and designing the CFSRE model. Comparative experiments against traditional full-sample learning methods (BiLSTM) and small-sample relation extraction methods (Proto, Siamese) demonstrated the effectiveness of our approach. Our method showed improvements in accuracy, recall, and F1 scores under various few-shot conditions, with significant enhancements observed in experiments like 5Way-1Shot and 10Way-1Shot on the DUIE Few datasets. Furthermore, ablation experiment results indicated that the data augmentation module based on large language models and the CFSRE module effectively enhanced model performance. Finally, data augmentation experiments confirmed that our proposed framework benefited the CFSRE model and provided gains for traditional methods. In the future, this approach will be applied to medical text information extraction, effectively reducing manpower costs, facilitating quicker understanding of patient information by doctors, and supporting the formulation of personalized rehabilitation treatment plans. In the future, we will combine the characteristics of medical texts with more specialized medical large language models, such as ChatMed and Zhongjing-LLaMA, during the process of enriching data samples. This will further optimize the performance of the method in the medical field and improve the efficiency of medical relation extraction.

REFERENCES

- [1] Wang C, Fan J. Medical relation extraction with manifold models[C]//Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). 2014: 828-838.
- [2] Peng C, Yang X, Chen A, et al. A study of generative large language model for medical research and healthcare[J]. NPJ Digital Medicine, 2023, 6(1): 210.
- [3] Wu X, Duan J, Pan Y, et al. Medical knowledge graph: Data sources, construction, reasoning, and applications[J]. Big Data Mining and Analytics, 2023, 6(2): 201-217.
- [4] Lu J, Gong P, Ye J, et al. A survey on machine learning from few samples[J]. Pattern Recognition, 2023, 139: 109480.
- [5] Han X, Zhu H, Yu P, et al. Fewrel: A large-scale supervised few-shot relation classification dataset with state-of-the-art evaluation[J]. arXiv preprint arXiv:1810.10147, 2018.
- [6] Gao T, Han X, Zhu H, et al. FewRel 2.0: Towards more challenging few-shot relation classification[J]. arXiv preprint arXiv:1910.07124, 2019.
- [7] Wu T, He S, Liu J, et al. A brief overview of ChatGPT: The history, status quo and potential future development[J]. IEEE/CAA Journal of Automatica Sinica, 2023, 10(5): 1122-1136.
- [8] Wei X, Cui X, Cheng N, et al. Zero-shot information extraction via chatting with chatgpt[J]. arXiv preprint arXiv:2302.10205, 2023.
- [9] Li S, He W, Shi Y, et al. Duie: A large-scale chinese dataset for information extraction[C]//Natural Language Processing and Chinese Computing: 8th CCF International Conference, NLPCC 2019, Dunhuang, China, October 9–14, 2019, Proceedings, Part II 8. Springer International Publishing, 2019: 791-800.
- [10] Xu J, Wen J, Sun X, et al. A discourse-level named entity recognition and relation extraction dataset for chinese literature text[J]. arXiv preprint arXiv:1711.07010, 2017.
- [11] Che W, Liu T, Li S. Automatic Extraction of Entity Relations[J]. Journal of Chinese Information Technology, 2005, 19(2): 1-6.
- [12] Zheng, Suncong, et al. "Joint entity and relation extraction based on a hybrid neural network." Neurocomputing 257 (2017): 59-66.
- [13] Wan, Zhen, et al. "Gpt-re: In-context learning for relation extraction using large language models." arXiv preprint arXiv:2305.02105 (2023).
- [14] Brinkmann A, Shraga R, Der R C, et al. Product Information Extraction using ChatGPT[J]. arXiv preprint arXiv:2306.14921, 2023.
- [15] Li G, Wang P, Ke W. Revisiting large language models as zero-shot relation extractors[J]. arXiv preprint arXiv:2310.05028, 2023.
- [16] Tianhao Li, et al. "A Chinese relational extraction model fusing ERNIE and attention mechanisms." Journal of Chinese Computer Systems 43(06) (2022): 1226-1231.
- [17] Ji Yimu, Zhang Wang, Liu Qiang, et al. Chinese text relationship extraction method based on small sample learning[J]. Journal of Nanjing University of Posts and Telecommunications: Natural Science Edition, 2023, 43(4): 64-71.
- [18] Lin, Chin-Yew. "Rouge: A package for automatic evaluation of summaries." Text summarization branches out. 2004.
- [19] Cui, Yiming, et al. "Pre-training with whole word masking for chinese bert." IEEE/ACM Transactions on Audio, Speech, and Language Processing 29 (2021): 3504-3514.
- [20] Soares, Livio Baldini, et al. "Matching the blanks: Distributional similarity for relation learning." arXiv preprint arXiv:1906.03158 (2019).
- [21] Ma, Yubo, et al. "Large language model is not a good few-shot information extractor, but a good reranker for hard samples!" arXiv preprint arXiv:2303.08559 (2023).